

Lec 24 Numerical Example

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Lecture	24 / 63 + intro
Duration	45:54
Captions	English (auto-generated)
Words	7,113
Video	0ui08gNfjA
Playlist	Fundamentals of Generative AI and LLMs

Auto-generated YouTube captions, lightly cleaned (removed [music] tags, grouped into paragraphs). These are **verbatim lecture captions**, not edited notes.

Transcript

[00:00] Hello everyone. Welcome to the lecture series on fundamentals of generative AI and the large language models. We are in the last session of week three. In this a particular week, we will not be seeing any hands-on, but we will be having a practical implementation of VAE in the coming session in the coming week as the first session. So so far, what is that we have looked into? We have seen the complete implementation, complete working mechanism of VAE. How exactly the encoder works, how decoder works, and what is the reparameterization trick. All that we have seen theoretically, mathematically. Now let us understand complete working mechanism with one numerical example. So in this particular session, we will understand

[00:47] the VAE working using a numerical example. So let us look into this particular problem. So what we are discussing is a complete numerical example, right? So the very first component in VAE will be the encoder part. So what we are looking into is how exactly the encoder part of VAE works. So here what we are seeing at this is an input. Right? And then what kind of input we have considered? Here we have considered a continuous data. So that is the data which is numerical, I mean continuous data. Now this data and more specifically we have taken a tabulated data.

So tabulated data is the data which is having in the in form in the form of rows and columns. And here there are four such features. Because

[01:40] there are four features, there are input neurons that we are replacing. What is the first step? The input has to be passed to a first hidden layer. So how is that the connection will be established? Is it not the first input will be passed to two every uh to the first neuron and the input will be passed to every of the neuron. But here the input is a pass to the first hidden neuron. So what exactly is going to happen here? There will be a summation and we will have an activation. So if at all if the summation has to happen, then is it not input will get multiplied with the weights and then we'll be adding the bias at also. So now how many connections are there? There are four such connections from first input to the

[02:32] first hidden neuron, second input to the first, third input to this and fourth. There are four such connections. So we are going to consider all of these weights. These are the four weights that will be placed on each of that connection and we have an bias at term and the bias value what we have used here is 0.1. Similar way, next we are going to establish a connection in this way. The input is a pass to two and second neuron in the hidden layer, right? So since there are four such connections, we are going to consider these four weights.

[03:15] These are the four weights. Similar way, we are going to establish a connection like this. So it is input is given to a third hidden neuron. So here once again we'll be using three you know, there are four such connections. So these are the four lay weights that we will be using. So I think the architectural diagram is clear. So we have an input that is a connected to a first hidden layer. So what is the operation that is going to happen? It is input getting multiplied with the weight and then the bias at term. So, this is what is the calculation that we are going to do. So, the first input 5.1 getting multiplied with this weight is 0.1. And the second input 3.5 getting multiplied with 0.4. The third input 1.4

[04:10] getting multiplied with -0.7. The weights are randomly assigned. When weights are randomly assigned, it can be positive value or it can be a negative value. So, here it is -0.7. And the fourth input getting multiplied with this weight is 0.2. And you're also adding the bias at term. So, the bias at term here it is 0.1. So, do a simple calculation math here. And then we have obtained each of these values. Now, what values we have obtained is the summation term. In each of this we got the summation term. Now, what is that we will be doing? We are going to pass an activation function, right? It is an hidden layer. If it is an hidden layer, what activation function we'll be using? Just recall from the concept. Is it not an

[05:00] non-linear activation function? So, here we are using an ReLU activation. So, ReLU is applied to the hidden layer. So, what is the ReLU formula? Is it not the max of 0 {comma} whatever is the input that you're passing? Here input what we

are passing is h . So, max of 0 {comma} h . So, now when I pass since it is 0 {comma} h , now when we pass the first value, h value is 1.07. So, when we pass 1.07, the maximum between these two is 1.07.

[05:39] So, h_1 value what we are getting is 1.07. So, that is this is a h_1 . And after we passed it to an activation function we are calling it as A . So and the H_2 this is a H_2 hidden neuron and it is 0, 1.93. The maximum between these two is 1.93. So it is 1.93. Then max of 0, 87. So maximum between these two is 0.87. So what is that we have completed in this particular slide is the input is a passed to the first hidden layer and here we have calculated the summation term and then we have passed it to an activation function and then we have obtained these values. So this is what is the final output of So this is A A_1 A_2 A_3 . So all this we are calling it as A and this is the output of the first hidden layer.

[06:41] Now once you complete the hidden layer now let us assume we have the next hidden layer. So what is that the next hidden layer? Let us assume here the latent dimension is two. What do you mean by latent dimension? It is we have one more layer where we have two neurons. Now if the latent dimension is two now just to recall from our theory concept what is that the encoder will give us? Is it not the encoder will output the appropriate posterior distribution parameters which means the encoder will output the mean and then the variance. If that is the case is it not now what we have obtained is the output of this particular layer. This is what we have obtained from our previous slide. Now from here what is that we

[07:35] have to get? We have to get the mean and variance. Now how many such mean and how many such variance we get? That depends on what is the latent dimension that we have to fixed. Because the latent dimension here is two. Therefore, we are going to get two you know, mean and two variants. How is that we calculate? Is it not the formula is once again a simple whatever is the output that we have obtained, that gets multiplied with the weight and there will be some bias set term. So, is it not we require some set of weights here also? The answer is yes and these are those set of weights. For a simple calculation, intentionally the bias is kept as a zero. Okay? So, now how is that the connection will be

[08:26] established? Is it not the connection is established in this way? We have an output and every of these values will be passed to the neuron. So, here we have three values and you are establishing a connection and then this is how the connection is been established. So, since there are three connections, is it not we require three weights and those three weights are these. Right? And similar way, is it not we establish a connection like this?

[08:59] Because ultimately we require two mean. So, if you require two mean, is it not you establish a connection like this and then we calculate? And here again we require three weights and these are the three set of weights. So, how is that we do a

calculation? Once again, it is a simple you know, multi-layer perceptron feed forward neural network. So, it is the values input getting multiplied with the weight and then the bias set term.

[09:27] So, is it not the calculation will be in this way? So, input is 1.07, right? That gets multiplied with the weight 0.5. Then the input 1.93 that gets multiplied with the weight -0.3 and then the input 0.87 gets multiplied with the weight 0.1. So, this is what we have completed and we also require a bias at home and that bias is 0.0. Right? Once again, you perform a calculation, you get this particular value. Similar way, you are establishing a second disconnection also, right? So, now what we have obtained is mean one. So, we have to now get this value that is mean two or μ two. So, how is that we get once again the input gets multiplied with the weights. Which weight? It is a these set of weights. So, input 1.07

[10:23] multiplied with the 0.2, 1.93 getting multiplied with 0.4 and 1.0.87 getting multiplied with -0.5 and then you will be adding the bias at home. So, after calculating, we get this particular value. Now, you can ask the question, "Ma'am, we have calculated this particular we have performed this calculation. Now, so far how did we do? Is it not the summation and then we had passed it to an activation function?" So, you can ask the question, "Are we not using an activation function here?" So, let us see what exactly is the answer. So, definitely this is what is the mean value we have obtained and your next question is this is what is the summation term. What about activation function? So, what is your

[11:14] answer? Can you think? I'll pause here for two seconds. Can you think whether we can have an activation function? What is your answer? So, answer is no, right? We cannot use an activation function. Can you tell why why we cannot use an activation function? Now, what is the activation function that we can use. Maximum, it's a non-linear activation function. So, suppose if you think of using an ReLU activation, so ReLU formula is max of 0, the input. Here, let us consider it as X. So, now what if this X value is negative? So, is it not you know, ReLU will convert it as a zero. Then, the mean value, the input max of 0, that X here is these are mean values.

[12:03] The mean value can be a positive value or it can be a negative value. Do you agree to this? Or it can be zero also. So, here what exactly? No standard activation functions will be applied when we are calculating this mean. The only answer The only reason why we will not apply is the mean can be either a negative value or it can be a positive value or it can be a zero. Now, when it is a negative value, that doesn't mean that it is wrong. The mean can take a negative value. But when you apply an activation function, so suppose if the mean is a negative and positive value, after we apply an activation function, is it not we get a zero and a 0.8? So, this value becomes a zero and 0.8 is a positive, it is retained.

[12:51] This is the reason why, you know, when we are after we get the output of first hidden layer, the next if you don't have any other hidden layers and if you wanted to get the latent dimension, that is when you have to get the mean and variance. Then, during the calculation of mean and variance, we will not be applying any hidden layer. I'm sorry, we will not be applying any activation function. Why we will not apply an activation function? Because if the mean is negative value, it is correct. We don't want to convert that into zero. That is the reason why we will not be applying an activation function. I hope this concept is clear.

[13:31] So, let me erase and then make it clear to you. Let us summarize what exactly we have done here. So, after we have got an output of the first hidden layer, the next step is we wanted to get the latent dimension. So, latent space. Before we get the latent space, we need to get the mean and the variance as an output. So, how is that we are calculating the mean? We are once again doing taking the whatever the output we have got, we are multiplying it with the weight, and we will add the bias at term. Only one important point that you need to remember here is we will not be applying any activation function. The reason why we don't apply an activation function here is if the mean is negative, we don't want to

[14:15] reduce that value. The mean can be a negative value. By applying an activation function, we are changing the negative value to positive. In order to avoid that, we are not applying any of the activation function here. Okay. So, what is the next step? From our previous slide, we have seen the mean is being calculated. What is the next step? We have to You remember from the theory that encoder will always give us the output mean and the variance. So, mean we have calculated. Now, this particular slide tells how to get the variance. So, how is that you do? Once again, whatever is the output that we have obtained, that will be passed.

[14:57] You'll be multiplying it with the weight, and then the bias at term. So, is it not This is the formula that you'll be using. So, [snorts] we get a log variance. So, from log variance, you will take the output which is denoted it as A, and you're multiplying it with the weight, and then the bias at term. So, here we are considering these set of parameters. These are randomly initialized. Over a period of time during back propagation, all of these parameters gets updated.

[15:27] That we will see. So, now how is that you do the connection? Is it not that the connection is established in this way? You are establishing it to the first neuron here and you will establish a connection to the second. So, when you establish a connection here to the first, you require three weights. Those three are these set of weights. And then you will be establishing this connection and we are considering these set of weights. So, how is that you do the calculation? Is it not that this is the calculation that we perform? You will take the input 1.07, multiply it with this weight.

[16:06] You will take this input 1.93. You will you know, multiply it with this weight.

You will take this input and multiply with this weight 0.2. Right? And then you will add the bias set up 0.0. So, » [snorts] » what is this 1.07? Is it not the output of the you know, whatever is the output that we have obtained from this? It is the same that we are considering even for the next also. So, we have these weights and then this is how you will be establishing the connection and we are calculating.

[16:41] So, once the calculation is done, this is what are the values that we have obtained. But the encoder is giving us the mean and a log variance. But ultimately, what is that we require? We require variance and from there we will be requiring a standard deviation. Right? So, sigma. So, is it not Now, from log variance, you have to convert it into variance. We have already seen what is the formula. It is e to the power of whatever is the value.

[17:13] So, the next step is since we have obtained this, now you have to convert it to log variance and you have to convert it into a standard deviation. So, how is that you get the variance first? The variance is obtained by using a formula e to the power of this log variance. So, e to the power of 0.732 » [snorts] » then we get this value. Then e to the power of 0.302, we get this particular value. Then, so what did we do? From a log variance we have converted into variance. But, what if you want a standard deviation? What is the formula? It is under root. So, now you will apply under root of this under root of this. We get this value and we get this value.

[18:04] What did we do? We have now calculated mean, we have calculated the log variance, from log variance we have obtained variance, and we have converted, not obtained, we have converted into variance and the standard deviation. Once again, here the concept will remain the same. Just because we have obtained summation, are we using an activation function? The answer is no. So, here no standard activation function is applied when to the layer, to this log variance layer of VAE. So, once again, it remains the same. We don't want to, you know, reduce the values. Whatever is the value that we have obtained, we want to retain as it is. So, no activation function has been applied.

[18:49] So, now what is that we have done so far? From input we have obtained the first hidden layer. From here we have obtained this mean and, you know, sigma. Now, what is the next step? Is it not you have to sample z? So far we have not sampled z. We have only calculated mean and log variance. From log variance, we have converted to variance and the standard deviation. Now, uh this is most important uh slide. Uh what is that we have to now do? We have to get Z latent space.

[19:27] From our theory, we have already seen how is that we get Z. Uh initially, we have understood Z is a sampled from this particular distribution. Uh whatever are the parameters that the encoder has produced. But, from reparameterization trick, we have understood that Z cannot be randomly sampled. Z is systematically um you know, obtained. So, how is that Z is obtained? Since there are uh size is two, you will

be uh obtaining Z_1 and Z_2 .

[20:02] Right? How is that you get Z_1 and Z_2 ? Z_1 is obtained by the formula mean one because here you are getting a mean one and a mean two. And here, we will be having uh standard deviation one and two or sigma one and two. So, now with the help of those two parameters, and you also remember reparameterization trick. What does it says? There is one random noise that is being added which is completely independent of this mean and uh sigma, μ and sigma. So, therefore, that is what is this term, uh constant term that is being uh Not the constant exactly. It is a sampled from normal distribution. So, the formula given for Z_1 and Z_2 is this.

[20:49] This we have seen in one of our previous sessions. Now, what is uh this value? Uh since this is a sampled from this a normal distribution whose a mean is zero and a standard deviation is one. So, uh now, you can randomly sample this value. Definitely, you'll be getting a different set of values, right? Now, how many such values you require? You require two such values. What is that? One is this and the second is this term. You require two such values. So, what you will do? Randomly, you will sample and you will take two values. So, those two values are these. One is 0.5 and second is -0.2.

[21:39] Why did we take these values? Because you want to calculate Z_1 and Z_2 . Why Z_1 ? Ultimately, you should get this is Z . But this has two features or two you know, yeah, two features. So, it is Z_1 and Z_2 . If you have to obtain Z_1 and Z_2 , the formula is you require two such values. So, you will be randomly picking one value. And again, you will randomly sample one more value. So, these two values. Now, simply substitute the values into this formula and you get Z_1 and Z_2 .

[22:16] What is this the first one? The first sample, I'm talking about this and in that we are calculating the first value. So, here μ this is μ_1 , μ_2 and this is variance one. I mean, standard sigma one, sigma two. So, you will take this value. Formula is mean this, then this is the formula, right? So, from this substitute, it is 0.043 and this is 1.442 and here this is 0.5. Do a simple calculation, you are getting 0.764.

[22:57] If there are any calculation typos, you can rectify uh you're also calculating along with me here. Now, similar way you have to calculate Z_2 . Z_2 is given by the formula mean two, standard deviation, and this is two. So, this is mean two volume, this is this value, and we are considering this. So, do a calculation, we are obtaining this value. So, now what we have obtained this is This is the value that we have obtained.

[23:32] Can you able to see? This is what we have obtained. Now, for the given input, we have seen this in theoretical theory concept very strongly. For a given input, you will have an n number of you know, n number of Z can be obtained. So, now for for the first for this particular combination, this is what we have obtained. Now, why did

we why is that we get a different even though mean and sigma is the same, but we have this particular term, right? Now, this term, next time when you sample it, you may get a different value from this. So, let us assume you have obtained this value.

[24:19] Next time when you sampled again for the same input, you have obtained this value. Once again, it is a sampling is a random operation, therefore it can be any value from this particular distribution. So, here we have obtained this. So, once again, substitute the values, so this is a mean one. Um So, when you substitute the values in this, we have obtained 0.043, and this is two. When you substitute the values in this, we get this. Now, what is that we have done? Is it a second input? No. For the same input, because of this is an random operation, now we have obtained this.

[25:05] For the same input, you there exist multiple such a samples. So, that is what I'm showing you here. For a different value of this gives a different sample the latent vector Z . And all is coming from the same distribution. What do you mean by this a same distribution? You see, this mean and standard deviation, we are not changing it. It is the same. How did we obtain that? How did we get this? We have obtained this from the input.

[25:34] Okay. Now, again you assume there is one more time you have sampled it. Now, this time you have obtained this value. So, um uh we have there is an extra bracket, okay. So, we have this is the value that we have obtained. Just substitute the values in this, and this is what is the calculation, and we are now getting this value. So, from this slide, what is that you are understanding? So, so far, input is passed to hidden layer. From hidden layer, we have obtained the mean and log variance. From vari- log variance to variance. From variance to standard deviation. From there, now you have to sample a Z . You have to obtain Z . Since the size of the Z is two, we require Z_1 and Z_2 . What is the formula to calculate

[26:26] Z_1 and Z_2 ? From reparameterization trick, we have already seen that this is Z_1 and this is Z_2 . So, now what is that we have to understand? For a same distribution, there exist n number of such a sampled latent vector Z . » [snorts] » So, for that, you are seeing these three example. So, I hope this is a clear to you. You have understood how to sample a Z . Now, what is the next step? Once Z is being sampled, the next step should be the first hidden layer at the decoder side. Now, here is the hidden layer at the decoder side.

[27:08] How is that? And you are saying that it is a hidden layer. When it is a hidden layer, there has to be an activation function. And we know that it has to be a non-linear activation function. So, it is a ReLU. We are using ReLU. How is that you will be establishing the connection? And what is Z that we have taken from our previous, we are considering this as our Z value. So, this is what you are seeing. This is the Z value that we have for taken.

[27:37] Right? Now, what is the connection you will establish? Is it not if this is your

Z value, that has to be passed as an input to the hidden layer? And hidden layer has a three neurons. So, now you have to establish a connection from every of the input, it has to go to the first hidden neuron. So, now you will be calculating thus. So, next you will establish a connection in this way. So, this is the second neuron at an hidden layer. You get this summation. Then you establish this connection and you get this term. So, and every how many such connections you are doing? You are doing two connections. So, so if you are required if you are doing two connections, is it not two weights that we get? Those

[28:26] weights are these. For every of this connection, two weights are this. And we are using a bias set up. And the bias is a small constant, 0.1. So, this is what is the output that we get for the first hidden neuron H1. And how is that you are calculating H1? H1 is 0.764. That is getting multiplied with its weight and its weight here is 0.6. Plus, you have a second value also and that second value is 0.318. That is getting multiplied with its associated weight. What is its associated weight? 0.3 and the bias term 0.1.

[29:08] Do a calculation. Because it is in a hidden layer, definitely it has to be passed through to an activation function. What activation function? Non-linear activation function and we have used ReLU. So, ReLU pass the H1 to ReLU, which is max of 0, H1. H1 is 0.6538. So, you get 0.6538. I hope the calculations are clear to you. Similar way, you now calculate for H2. So, what is the H2? So, you are now considering these set of weights. So, input 0.764 getting multiplied with its weights -0.4 plus 0.318 getting multiplied with its weight 0.1 plus the bias term 0.1.

[29:53] Do the calculation and then pass it to do an ReLU and ReLU will give max of 0, some value. Because here the value is a negative, max of 0, negative is 0. So, we have obtained a 0. Similar way, we are left out with a hidden layer hidden neuron H3. So, here is the calculation for H3. Once again, it is the same process. Take the input getting multiplied with its associated weight. 0.2 and -0.5. And then there is a bias term 0.1.

[30:29] Perform the calculation. Pass it to ReLU because it is a positive, you have retained it. So, what is that we have obtained? So, we have now obtained the output of the first hidden layer in the decoder part. So, since this is a decoder hidden layer, the activation function has to be applied and that activation function here is a ReLU. So, after [snorts] we applied ReLU, here it is A1, A2, and A3, which is denoted as vector A and this is the value that we have obtained, right? So, I hope this is clear. What we have done is the decoder part first hidden layer.

[31:10] What is the next step? From here, you have to go to this layer. This is the reconstruction one. Now, let us look into that part. Now, since this is the decoder output layer, what we are discussing is this particular layer. This is the decoder output layer, which means you have to use an activation function. Now, what activation

function that you'll be using here, that primarily depends on what kind of data that we are generating. Now, here, since recall the input what we have considered here is a continuous value, then output also should be a continuous value. If output is a continuous value, then what should be the activation function that we'll be using? What is the activation? Right, linear activation function, right? So,

[31:58] what activation function that we use here is linear activation function. So, you have a function to that some value will be passed and the output will be whatever is the value that we are passing. So, now, what if the input is binary or if the values are scaled between 0 and 1 or 0 1. So, what if we have a normalized between 0 and 1 or if it is a binary data, then we'll be using a sigmoid activation function. So, this is what is the sigmoid. So, you will apply sigmoid and the you know, summation term. So, now let's get it started.

[32:39] What is that we are doing? We are doing a from three to this a four. This is what from hidden neuron three to and a reconstruction four neurons. So, first let us see this X_1 you know, hat. So, how is that you will be calculating? Uh so, is it not the connection is established in this way? So, let's see. So, from this particular input, you are passing you are establishing a connection to this neuron. So, when you establish a connection, since there are three such inputs that are passed, you require the three connections and the three weights.

[33:18] Those are three weights are these: 0.2, 0.1, and -0.3. So, and the similar way, you will be establishing a connection for this. Again, you are requiring three connections, three such neuron weights. So, this is what is the calculation. So, input what we have considered is the output of our previous one, so which is 0.6538, that is getting multiplied with its weight 0.2. And for the second neuron, the output was zero. Zero getting multiplied with the 0.1 and the 0.0938 getting multiplied with this. What are these values? These values are the outcome output of this neuron, this hidden layer. So, that gets multiplied and you are obtaining 0.10 262. And what you have obtained is this a summation term. Now, you have to pass

[34:16] it to an activation function, and because the input is considered as a continuous value, we are using a linear activation function. What is linear? Even if you pass this value to a linear activation function, you ultimately get the same value. We obtain this. So, similar way for the second X_1 hat, so what you are now getting is X_2 hat. So, you will take the same input from this particular layer. You will now multiply it with its weight. And which are those weights? It is these set of weights. And then this is the value that we have obtained.

[34:53] Since it is a linear activation of function, once again you are passing it to a linear activation of function, and the output will remain the same. This is what is the output that we have obtained. So, this is what is the two In the In this particular slide, we are seeing the output of next two neurons. So, this is X_3 hat, and we have This is

X4 hat. You will do these two calculations. So, I hope the calculation part is clear. So, now what is that we have obtained? Is it not we got the output of the decoder output layer? At the decoder side, this is the output layer. And we got the output. By end of this, we got X1 hat, X2 hat, X3 hat, and X4 hat. Is it not Whatever we have obtained is the reconstructed value?

[35:47] That's the ultimate aim of variational autoencoder. It has to generate or reconstruct the data. Now, what we have obtained is the value, reconstructed value, or generated value. Now, so the final reconstructed output, is it not this is the final reconstructed output? Since the values are too many after the decimal, let us round off to you know, approximately three decimal. So, after rounding off, this is what is the value that we have obtained. How did we round off? Because it is a five after decimal since there are five digits, we have considered three digits.

[36:22] Now, whatever is the output that we have obtained from the decoder side. So, that is nothing but you remember go back from our go back to our theory concept decoder will always give us the distribution, right? And Z is a sample from that distribution. This is what theory says. But what does the practical concept says? The practical says whatever is the output that we have obtained, that itself that is the mean, that itself is the answer. So, now what we have obtained is X1 X2 X3 X4 hat, which is nothing but the mean value.

[37:02] Okay? So, the mean value is this. Now, you can see the original input is this and the reconstructed input reconstructed value what we have obtained or generated data what we have obtained is this. Is it not we have a huge difference? Why is that it is a huge difference? Because reconstruction here is very poor because whatever the weights and biases that we have considered are all randomly initialized. But after once the training is over, during back propagation, is it not we update all of these weights and all of these bias? Then over a period of time, is it not this value will be very much closer to reconstruct original value?

[37:46] Do you agree to this? So, this is what I have explained. When you during theoretical concept we say that Z is sampled from this particular distribution because decoder has to output the parameters and Z has to be reconstructed value has to be sampled from this distribution. This is what the theory we say. But in the practice or from during this particular numerical example, what did we understand? That whatever is the reconstructed value, that is you know, the mean whatever we have obtained, that itself is the reconstructed value. So, this is what is the decoder part. What did we understand so far? We have taken an input, we have passed it to an encoder, then we from encoder we have output we have obtained

[38:34] the mean and log variance. We have converted it into variance standard deviation, then we have sampled a Z. How did we get a Z? During I mean using a reparameterization trick formula. Then you have passed a Z to $2n$ and we have seen

that for the same input there exist a multiple such values. Multiple Z sampled can be you can get multiple Z . Now, why? Because you have you are taking the constant that is you know, from a standard deviation from that particular distribution we are taking those values.

[39:13] Now, once you complete once you get a Z , you are passing it to an decoder. Here at a decoder we have an hidden layer and we have an output layer. Because it is an hidden layer you have used a non-linear activation function. When it comes to an output layer, you have to use an activation function, but it depends on what kind of data that we are considering. Suppose if it is an continuous data linear activation. If it is a binary or normalized to zero and one, it will be a sigmoid activation.

[39:42] And whatever the output we have obtained from a decoder, that itself is the reconstructed or the generated data. Okay? What is the next step? You have to calculate the loss. What is the loss? Because we have considered a continuous value. Remember, the loss term should be should be mean squared error. Mean squared error. So, this is what is the mean squared error. And the mean squared error you will compare between X and $\hat{X}$ and here $\hat{X}$ is nothing but the mean value only. So, now you will consider this is your uh um original input X and the reconstructed or generated one $\hat{X}$ is nothing but the mean and this is the value.

[40:29] Now, what values we have considered? That rounded off value. Now, simply substitute the values into this formula and then do a math and we are going to get the reconstructed value as our reconstructed loss as 9.7111. So, how did we obtain this? It is mean squared error. It is between original and then the reconstructed one $\hat{X}$. So, just substitute the values in each of these. We get substitute the values and then we obtain reconstruction loss in this way.

[41:06] Next, what is the the variational VAE loss? We have constructed a we have obtained the first term reconstruction loss, but we have a second term as well. What is the second term? KL divergence. What is KL divergence responsible for? Is it not from theory we have understood that encoder distribution should remain closer to a prior distribution. So, this is encoder and this is the prior distribution. So, in a closed form, that is what do you mean by closed to form?

[41:38] We want some equation where we can substitute some values and get the answer. So, in a closed form, KL divergence when two distributions are Gaussian, we we get this particular equation. So, from our all our calculation, we have obtained the mean value, we have log variance, we have variance. Just get all those values and substitute in this. So, can we see? This is the mean value that we have obtained and this is the log variance we had and from log variance we had converted into variance.

[42:12] So, this is all the values. Just to go back from our previous slides. Somewhere in third or fourth slide we have done this calculation, get those values. Then, in this particular formula, we require the square mean square we require. So, just compute

it. So, how is that you compute? Just take the mean value and then perform a square of it and this is the value that we are obtaining. Now, once this is done, is it not a simply substituting the values into that KL divergence formula? So, substitute the values. How is that the value?

[42:46] First is a mean So, you are taking this. Then, it is you know, since there are two values. So, you're considering the first one. So, this is one, this is » [clears throat] » two. So, you're considering this. Then, you have minus log variance. What is log variance? Is it not This is the log variance. You have to consider minus one. So, and then you have second set of values. So, all those the second set values are these. So, perform a simple calculation, perform addition, subtraction, add up, and then finally divide it with the two, you will get this value.

[43:29] Since it is, you know, 0.3517, you're rounding it up 0.352. Now, you have in the previous slide you have computed a reconstruction loss. In this particular slide you have constructed a KL divergence. What is the total loss? Is it not the addition of those two? So, therefore the total loss is a reconstruction loss and then the KL divergence term. When you add, you are getting this. The whole intention and the total loss here is a 10.0631.

[44:01] The whole intention of this particular session is because VAE is one of the important concept, there [snorts] are so many terms. So, probably the reader will be having some doubts. How is that your sampling? How is that Z1 is obtained? How is that Z2 is obtained? So far, we have seen only theoretically. So, in order to understand the overall flow thoroughly, we have taken one numerical example. And this numerical example only shows you the forward pass from input to encoder to latent space to and decoder, and then we have computed loss. What is the next step? Because the loss value is very high. Where the it is a 10.0631.

[44:48] So, because the loss is very high, what is the next step? We have a role of an optimizers. So, we have optimizers who will come into the play. And then, during the back propagation, you will update each of these parameters. By updating the parameters over a period of time, is it not all the weights are optimized? Is it not biases optimized? With that, is it not we get better mean and then the log variance? Then from there, we get a better Z. From there, we get the better reconstructed one or regenerated one.

[45:22] So, this is the overall working mechanism of the VAE. And with this, we have come to an end of this particular week three, which has completely focused on a variational autoencoder. So, in our next week, that is in week four, the first session will be the practical implementation of VAE. And then, we will see the further theoretical concept in week four. Thank you.